

Review of Current Rake Planning Sheet

Overall Assessment

Your team has already built a fairly strong operational control sheet. The workbook covers:

- Rake positioning
- Pendency visibility
- Inventory
- Wagon status
- Route tracking
- Loading status
- Forecast vs Actual tracking
- Restrictions and operational constraints

This is operationally useful, but the current structure is still heavily dependent on manual monitoring and reactive planning.

The biggest opportunity is to move from:

"Tracking operations" → "Predicting and controlling operations"

Key Gaps Identified

1. Too Much Information on One Summary Screen

The Summary tab mixes:

- Pendency
- Rake movement
- TXR dates
- ETA
- TAT
- Planning remarks
- Inventory details

This creates cognitive overload.

Impact

Controllers and planners may miss:

- Late rake risks
- TXR expiries
- Empty repositioning opportunities
- Priority customer allocations

2. Planning is Mostly Static

The current file appears to show:

- Current rake position
- Existing plan
- Current loading status

But it does not clearly show:

- Predicted future shortages
- Expected rake availability by time bucket
- Priority-wise rake allocation
- Dynamic rerouting recommendations

This means planners still depend on manual judgement.

3. No Exception-Based Planning Layer

Currently every rake gets equal visibility.

Operational teams should focus only on exceptions such as:

- TAT breach risk
- ETA delays
- TXR expiry within next 48 hrs
- High pendency locations
- Unproductive idle rake time
- Route congestion

Without exception highlighting, the team spends time scanning instead of deciding.

4. Forecast vs Actual is Not Integrated into Live Planning

You already maintain a Forecast vs Actual sheet.

But the forecast is not visibly driving:

- Rake prioritization
- Empty rake repositioning
- Loading sequence
- Inventory balancing

This is a major missed optimization opportunity.

Recommended Improved Planning Structure

A. Create 3 Separate Planning Layers

Layer 1 — Control Tower (Live Operations)

Purpose: Real-time monitoring.

Should contain only:

Metric	Purpose
Rake ID	Tracking
Current Location	Visibility
ETA	Arrival prediction
Delay Status	Exception handling
TXR Due	Maintenance risk
Next Loading Point	Operational clarity
Priority	Allocation logic
Risk Indicator	Decision making

Use traffic-light color coding:

- Red → action required
- Amber → monitor
- Green → healthy

Layer 2 — Planning Engine (72 Hr / 7 Day)

Purpose: Future allocation planning.

Should include:

Planning Item	Example
Expected rake arrivals	Next 72 hrs
Forecast loading demand	Customer-wise
Empty availability	Base-wise
Loading capacity	Terminal-wise
Route restrictions	Congestion / blocks
Priority allocation	Revenue/customer-based

This becomes your real decision-making sheet.

Layer 3 — Performance Dashboard

Purpose: Management review.

KPIs should include:

KPI	Why Important
Average TAT	Asset utilization
Idle rake hours	Productivity loss
TXR overdue count	Compliance
Loading adherence	Execution quality
Forecast accuracy	Planning maturity
Empty movement %	Cost efficiency
Pendency trend	Customer service

Specific Improvements Recommended

1. Add a Rake Priority Score

Currently planning seems manual.

Introduce a calculated score using:

Factor	Weight
Pendency severity	30%
Customer priority	25%
TXR urgency	15%
Distance to terminal	15%
Idle time	15%

Then auto-rank rakes.

This removes subjective allocation.

2. Introduce Predictive ETA Risk

Instead of only showing ETA:

Add:

- Expected delay probability
- Congested route alerts
- Historical route TAT variance

Example:

Rake	Planned ETA	Risk
ADANI-105	25 May 18:00	HIGH
RK-46	25 May 08:35	LOW

This enables proactive planning.

3. Add Auto Alerts

Your file should automatically flag:

- TXR expiry within 2 days
- Rakes idle > 12 hrs
- TAT above benchmark
- Pendency above threshold
- Inventory below minimum

These should appear automatically on top.

4. Convert Planning from Rake-Centric to Demand-Centric

Current structure appears rake-focused.

Instead, first calculate:

- Which location needs capacity most urgently
- Which customer has loading commitment risk
- Which route has best turnaround

Then assign rakes.

This improves utilization.

5. Create a Rolling Allocation Model

Instead of static daily planning:

Maintain:

- T+0 (today)
- T+1
- T+2
- T+3
- T+7

This helps avoid tomorrow's shortages.

Structural Excel Improvements

Recommended Automation

Use Power Query For:

- Live rake imports
- Route status updates
- Inventory refresh
- Wagon status integration

Use Power Pivot / Data Model For:

- KPI dashboards
- Trend analysis
- Forecast comparison

Use Conditional Formatting For:

- Delay risks
- TXR alerts
- High pendency
- Empty imbalance

Use Slicers For:

- Base
 - Route
 - Customer
 - Rake type
 - Delay category
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Operational Improvements

Suggested Planning Cycle

Every 4 Hours

1. Refresh rake positions
 2. Check ETA variance
 3. Recalculate priority score
 4. Reassign empties if required
 5. Publish updated 24 hr plan
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Daily Planning Meeting Structure

Section 1 — Exceptions (10 mins)

Only discuss:

- Delayed rakes
- TXR issues
- Congestion
- Critical pendency

Section 2 — Next 72 Hr Capacity (10 mins)

Review:

- Expected arrivals
- Expected loading demand
- Terminal constraints

Section 3 — Reallocation Decisions (10 mins)

Decide:

- Diversions
 - Empty repositioning
 - Priority customers
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Biggest Opportunity

The workbook currently behaves like:

"A detailed operational register"

It should evolve into:

"A predictive rake allocation control tower"

That single shift will:

- Reduce idle time
 - Improve TAT
 - Reduce emergency planning
 - Improve customer commitments
 - Improve asset utilization
 - Reduce planner dependency on manual tracking
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Final Recommendation

If you want maximum improvement, the best next step would be:

Phase 1

Clean and simplify the operational dashboard.

Phase 2

Build a predictive 72-hour planning engine.

Phase 3

Automate exception alerts and priority scoring.

Phase 4

Move the entire model into Power BI or a live control tower dashboard.

That would significantly improve planning quality versus the current Excel-driven operational tracking model.